Day 09: Linux Basics

Muhammad Usman
Cybersecurity Learning Log

Part 1: Linux Theory

1. What is Linux?
Linux is an open-source operating system kernel. It is used in many operating systems, servers, computers, and cybersecurity environments.

2. What is a Linux distribution?
A Linux distribution is a complete operating system built around the Linux kernel, usually including system utilities, software packages, and tools.

3. Name four Linux distributions.
Four examples are Ubuntu, Debian, Fedora, and Kali Linux.

4. What is Kali Linux?
Kali Linux is a Debian-based Linux distribution designed for penetration testing, digital forensics, security assessments, and other cybersecurity work.

5. What is a terminal?
A terminal is an interface where users enter text commands to interact with the operating system.

6. What does the whoami command do?
It displays the username of the account currently running the command.

7. What does hostname display?
It displays the system's hostname, which identifies the computer on a network.

8. What does pwd mean?
pwd stands for Print Working Directory. It shows the full path of the directory I'm currently in.

9. What is the purpose of ls?
The ls command lists files and directories in a specified location or the current directory.

10. Explain the difference between ls, ls -l, and ls -a.
ls displays a basic listing. ls -l shows a detailed listing with information such as permissions, ownership, size, and modification time. ls -a also displays hidden entries, including names beginning with a dot.

11. What does cd do?
The cd command changes the current working directory.

12. What does cd .. do?
It moves to the parent directory of the current directory.

13. What does cd ~ do?
It takes the user to their home directory.

14. What does cd / do?
It changes the current directory to /, the root of the Linux filesystem.

15. What is the Linux root directory?
The root directory, represented by /, is the top-level directory of the Linux filesystem. Other directories and files are organized beneath it.

16. What is the difference between the root directory and the root user?
The root directory / is a filesystem location. The root user is the administrator account, which normally has extensive privileges over the system.

17. What is an absolute path?
An absolute path specifies a location starting from the root directory /, such as /home/user/Documents.

18. What is a relative path?

A relative path describes a location in relation to the current working directory rather than starting from /.

19. How are hidden Linux files commonly identified?

Hidden files and directories commonly have names beginning with a dot, such as .config. They are shown by commands such as ls -a.

20. Why are Linux skills important for cybersecurity?

Linux skills help cybersecurity professionals navigate systems, inspect logs, understand permissions, use security tools, and investigate system or network activity.

Part 2: Practical Screenshots (Kali Linux)

The following screenshots were taken in the Kali Linux terminal and show the basic navigation and identification commands.

Figure 1: whoami



The current user is **kali-linux**.

Figure 2: hostname



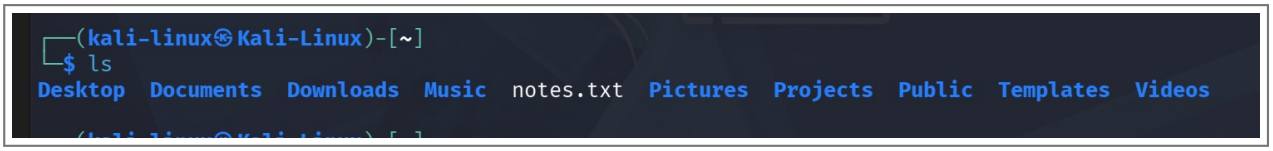
The hostname of the system is **Kali-Linux**.

Figure 3: pwd



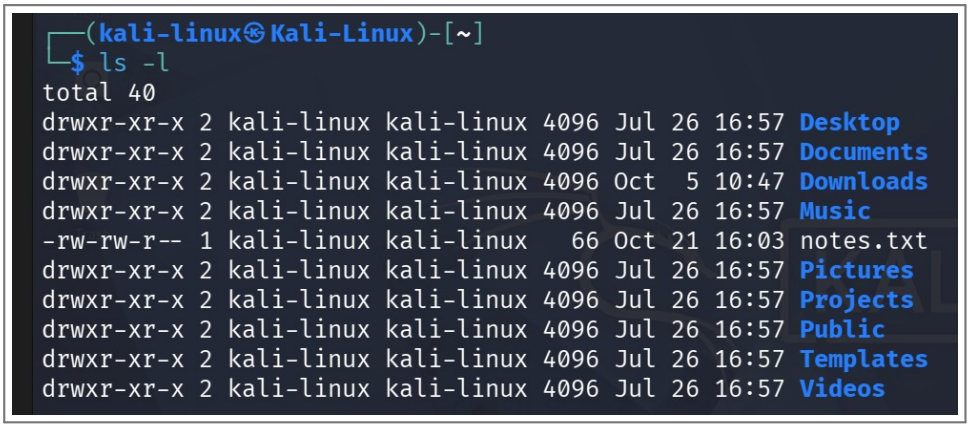
The current working directory is **/home/kali-linux**, the home directory of the user.

Figure 4: ls



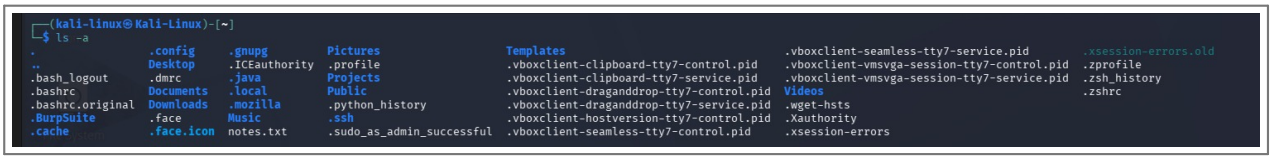
A basic listing of the home directory: Desktop, Documents, Downloads, Music, notes.txt, Pictures, Projects, Public, Templates and Videos.

Figure 5: ls -l



A detailed listing showing permissions, owner, group, size and modification date. Directories start with **d** (for example drwxr-xr-x), and **notes.txt** is a regular file (-rw-rw-r--) of 66 bytes.

Figure 6: ls -a



Lists all entries including hidden ones whose names begin with a dot, such as .bashrc, .config, .ssh and .zshrc, along with the . and .. entries.